

Appendix 1

Table 1: The results of open coding analysis

ID	Evidence excerpt (raw statement)	Initial concepts	Subcategory
1	SN213 Scot Explorer's master was undertaking other duties on the bridge and was not monitoring nearby traffic.	IC01 Failure to maintain a proper lookout	IS ₀₁ : Lookout fault
	SN25 A proper and effective lookout was not maintained on board either vessel. Good Intent's wheelhouse was left unattended for long periods of time and Silver Dee's skipper was not monitoring nearby vessels.	IC02 Failure to maintain regular lookout	
	SN37 Phoenix's owner did not see the ferry approaching on the motor cruiser's port quarter because he was looking ahead.		
2	SN85 The VHF conversation between Huayang Endeavour and Seafrontier was confusing, did not use SMCP terminology, and resulted in the two bridge teams developing conflicting plans while believing that they understood the intentions of the other.	IC03 VHF communication not maintained on effective duty	IS ₀₂ : Inadequate use of monitoring tools
	SN210 Hannah Lee's radar failed shortly after departing Mandurah; the skipper did not attempt to repair it or use alternative aids (e.g., VHF channel 16) to detect surrounding vessels.	IC04 Navigation aids not used	
	SN338 The Century Queen and Kaytlin Marie failed to confirm a passing arrangement via VHF early: the downbound Kaytlin Marie did not propose the passing method, and the Century Queen did not verify the intention timely, leading to a collision at mile 126 of the Lower Mississippi River.	IC05 Lack of communication with overtaking vessel via VHF	
	SN402 The container ship Ocean Voyager and the tugboat Sea Master collided near the entrance to the port of Charleston. The two vessels failed to establish early and effective VHF communication to confirm a passing arrangement.		
	SN18 Both Electronic Chart Display and Information System (ECDIS) units were set to silent mode, with all audible alarms deactivated while underway.	IC06 Ineffective use of radar and other navigational AIDS	
	SN53 The OOW of Saga Sky (passenger ship) did not use radar plotting effectively during night navigation	IC07 No radar plotting performed	
	SN105 The Maltese bulk carrier Malta Bulk overloaded by 3,000 tons, with reduced stability due to overload.	IC08 Reduced stability due to overload	
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3	SN87 The Maltese bulk carrier Malta Pride transported iron ore off the northwest coast of Australia with an actual cargo weight of 45,000 tons, exceeding the approved deadweight of 40,000 tons, reducing the vessel's stability.	IC09 Improper loading of ship cargo	
	SN113 The vessel exceeded the allowable limit, and improper stowage and lashing of heavy cargo shifted the center of gravity to port, degrading stability and maneuverability during navigation and thereby increasing collision risk.		
4	SN338 The Canadian container ship Canadian Express loaded containers at the Port of Vancouver; it stacked 18 heavy containers (30 tons each) in columns 2-4 on the port deck and concentrated light containers on the starboard side.	IC10 Cargo stacking not balanced	IS ₀₄ : Uneven cargo loading and inadequate securing
	SN113 The Chinese general cargo ship Hua Yang 7, when loading mechanical equipment, did not secure heavy equipment (15 tons per unit) in the middle of the cargo hold as required by the loading manual, but only simply lashed it to the port area, causing the vessel's center of gravity to shift to port. During navigation.		
5	SN401 Fishing vessel Nippon Maru had an approved navigation area of the Sea of Japan inshore (≤ 60 nautical miles from the coast). The skipper, to increase catches, unauthorizedly sailed to the Alaskan waters (U.S. EEZ, 220 nautical miles from the coast), sailing beyond the approved navigation area.	IC11 Sailing beyond the approved navigation area	IS ₀₅ : Sailing beyond the navigation zone
	SN301 general cargo ship Aussie Star (draft 5.1 meters), without applying for a permit, unauthorizedly entered the Yangtze estuary deep-water channel (only for large vessels with draft ≥ 10 meters), making unauthorized entry into the deep-water channel.	IC12 Intrusion into the deep-water channel of the Yangtze estuary	
	SN384 The Hammersmith Bridge encroached toward the channel center.	IC13 Channel deviation	
	SN503 The British aquaculture vessel UK Aqua, to inspect offshore aquaculture cages, unauthorizedly entered the restricted warning area of the Great Barrier Reef (temporarily closed for coral protection), navigating in the restricted/warning area.	IC14 Navigation in restricted or warning areas	
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	SN18 The containership Maersk Wisconsin agreed to a “port-to-port passing” with the tug-barge unit Ruth M. Reinauer in the Kill Van Kull channel (1150 feet wide), but failed to maintain its channel position and encroached into the outbound lane.	IC15 Encroachment of outlet lane	
6	SN27 Petunia Seaways had a speed over the ground of 14 knots and was not sounding a regular fog signal.	IC16 Failure to implement fog navigation measures in accordance with regulations and company safety management requirements	IS ₀₆ : Regulatory and procedural non compliance
	SN125 The OOD of the USCG Cutter Thetis did not activate the radar automatic alarm function, failed to detect the approaching tugboat Matachin from the stern in time, leading to collision.	IC17 Failure to use radar automatic alarm and other safety navigation equipment	
	SN283 The Capt. Shorty C and Jackie operators met in a section (Bolivar Buoy 15-20) with strong currents/shoaling, contrary to Coast Pilot 5’s guidance.	IC18 Non-compliance with traffic control regulations	
7	SN116 Norma Jean increased speed to 20 knots after passing the channel entrance beacon, with the skipper failing to reduce speed in near-total darkness (moonrise at 0616, collision at 0625) and unknown anchorage areas ahead.	IC19 Not sailing at a safe speed	IS ₀₇ : Didn’t maintain a safe speed
	SN235 The Mark E Kuebler approached the Nisalah at 11.6 knots (exceeding safety speed), was sucked in by hydrodynamic forces.	IC20 Failure to adopt an appropriate safe speed for the prevailing environment and conditions	
8	SN305 Chester’s Anchor light used Narva 21/5W 12V bulb (incorrect for 10W lens), jammed in socket with insulating tape; visibility reduced to <1 mile (required ≥2 miles per COLREGS).	IC21 Failure to display anchor signals after anchoring	IS ₀₈ : Signal display negligence
	SN406 Neither vessel made sound signals, losing the opportunity to alert the other to their intentions.	IC22 Overtaking signal not used in the channel	
	SN317 Stella VII displayed “red over white” fishing lights (not engaged in fishing), confusing Bunga Orkid Tiga’s OOW	IC23 Signal light not displayed correctly as per regulations	
	SN409 Marcliff (give-way vessel) should alter to starboard to avoid crossing ahead; instead turned port. APL Guam (stand-on) ordered hard starboard but too late.	IC24 Left turn accelerating imminent danger	IS ₀₉ : Improper avoidance measures
	SN239 The Monte Alegre pilot increased speed from slow ahead to half ahead without informing the Chembulk Houston deputy pilot, reducing the speed differential and extending the overtaking time.	IC25 Avoidance measures not coordinated	
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9	SN340 Silver Yang’s second mate, after identifying Ella’s Pink Lady (on the port beam, 0.6 miles away, a beam vessel), initially ordered “20° starboard helm” but then incorrectly changed to “hard a port” 10 seconds later, causing the ship to turn left toward the beam yacht.	IC26 Left turn directed towards the vessel ahead	
	SN405 Hannah Lee’s skipper, realizing collision was imminent, swung the helm hard to port (contrary to COLREGS Rule 15 which requires give-way vessels to alter course to starboard in crossing situations), increasing the impact angle.	IC27 Improper avoidance measures	
	SN431 When the heavy lift vessel Hawk was turning, the pilot did not take the most effective collision-avoidance action of “flanking” and still operated in the conventional turning mode, leading to collision with the moored boat Desakata.	IC28 Failure to take actions that would most effectively avoid a collision	
	SN335 The Bow Fortune’s pilot hailed the Pappy’s Pride 3 times and sounded 5 short blasts (danger signal) but took evasive action (starboard turn, speed reduction) too late.	IC29 Failure to take necessary actions to help avoid a collision	
10	SN302 When the towing vessel St. Rita moved the barge in strong current (5-6 knots), it did not pre-evaluate the impact of the current on the barge and did not formulate an emergency avoidance plan, leading to the barge losing control, colliding with a group of moored barges, and sinking.	IC30 Failure to take effective preventive and emergency measures	
	SN475 After the main engine failure of the oil tanker FR8 Pride, the pilot ordered hard starboard rudder to avoid the MODU Rowan EXL I; the tugboats towing the drilling unit failed to adjust the towing direction in time, and both vessels took improper emergency maneuvers.	IC31 Both ships failed to execute proper emergency maneuvers	IS ₁₀ : The emergency response was inadequate/untimely
	SN424 After the fishing vessel Kathleen K lost engine control, the captain did not activate the emergency stop procedure first but left the bridge for the engine room, leading to the vessel losing control and colliding with moored vessels, with inadequate emergency response measures.	IC32 Improper emergency measures	
	SN37 The St. Rita’s captain had limited fleeting operation experience	IC33 Lack of experience in avoiding fishing boats and maneuvering ships	
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ID	Evidence excerpt (raw statement)	Initial concepts	Subcategory
11	SN49 The OOD-2 of the submarine Greenville was unfamiliar with the operation of the sonar repeater (AVSDU); after the equipment failed, he did not check sonar room data frequently as required, failing to detect the fishing vessel Ehime Maru in time.	IC34 The second officer is not familiar with the ship's navigational aids	IS ₁₁ : Inexperience
	SN221 The captain of the towing vessel St. Rita lacked experience in moving barges with a single line in strong current, misjudged the turning timing, leading to the barge being pushed by the current into the group of moored barges.	IC35 Captain's inexperience	
12	SN153 The trawler Jay Dee sank rapidly after being hit by an unidentified trading ship; the trading ship did not stop to assist, did not respond to VHF calls, and sailed away directly, constituting a hit-and-run after the collision.	IC36 Hit-and-run incident	IS ₁₂ : Failure to correct errors in time
	SN243 Spartia continued to Bunbury without stopping after collision	IC37 Drove off the scene after the accident	
	SN286 An unmarked Maltese cargo ship illegally transported 60 tons of crude oil without legal procedures off the Australian coast. It did not activate AIS, turned off navigation lights, and loaded/unloaded cargo at night. Illegal maritime transport was not detected or prevented.	IC38 Ship's illegal operations not detected and stopped in time	
13	SN120 the submarine's command team assessed that Karen was a merchant vessel, primarily because no trawl noise was heard. The submarine was at a depth where it could, if necessary, pass safely beneath a merchant vessel, therefore the command team would not have perceived any risk of collision; as a result, no avoiding action was taken.	IC39 Failure to adequately assess the ship's environment when avoiding another vessel	IS ₁₃ : Lack of safety awareness
	SN120 The SANDY RICKMERS sailed through the siding's stop signal and failed to maintain position, violating the Kiel Canal's traffic signal rules.	IC40 Failure to implement required safe navigation measures	
	SN58 The second officer was distracted throughout his watch by the continual use of a tablet computer.	IC41 Engaged in non-navigation related work during the duty period	
	SN127 Big D pilot proposed 'one whistle' (port-to-port), Carol McManus pilot agreed but later maneuvered toward the right descending bank (Big D's side).	IC42 Driving without due care	
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14	SN114 The fishing vessel Black Sheep blindly crossed ahead of the containership Columbus Canada in the Galveston Bay Entrance Channel without confirming the containership's movement, leading to collision.	IC43 Blindly crossing the bow of another ship	IS ₁₄ : Unsafe decision making under perceived risk
	SN108 The Seeley's mate performed a "tripping maneuver" (released starboard lines between Weeks 207 and Weeks 213) in a narrow channel (130ft wide) near Hinckley Marina, ignoring the nearby turning basin.	IC44 Failure to exercise sufficient caution while navigating in narrow waterways	
	SN124 The towing vessel Natures Way Commander attempted to emergency stop to avoid moored barges in heavy rain (34-knot winds), but blindly reversed without assessing current impact, leading to the tow losing control.	IC45 Blindly taking evasive actions	
15	SN102 the skipper on board Guiding Light had left the wheelhouse unattended while the two vessels were transferring the cod end and did not perceive the risk of leaving the wheelhouse while in close proximity to the partner vessel.	IC46 Captain's failure to effectively discharge supervision and guidance duties	IS ₁₅ : Crew/captain unqualified/unfit
	SN38 The bosun's knowledge of radar and navigation was rudimentary and that he had no understanding of the COLREGS.	IC47 Incompetency of crew	
	SN53 Vision's skipper had not completed the PPR course and his RYA Yachtmaster qualification had not been commercially endorsed.	IC48 Ineligibility of the driving duty officer	
	SN39 The helmsman was inexperienced at steering the vessel into Cowes Harbour and was insufficiently practised at steering by digital compass alone.	IC49 Driver incompetence	
16	SN139 The containership Hammersmith Bridge was undermanned during watch, with only 1 officer (chief mate) and a helmsman on the bridge while the master rested; insufficient manning prevented effective monitoring of channel dynamics, leading to collision with the St. Louis Express	IC50 Ship undermanned	IS ₁₆ : The manning was insufficient
	SN102 The U.S. tug US Tug had only 2 crew on watch (lacking dedicated lookout), not equipped with adequate crew; it failed to detect British fishing vessel UK Fish, bow colliding.	IC51 Insufficient crew on board	
	SN348 The absence of a dedicated lookout on Scot Explorer's bridge meant there was no one immediately available on the bridge to assist the master as the situation developed by activating hand steering and taking the helm.	IC52 Lack of dedicated lookout personnel leading to collision risk with fishing boats	
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ID	Evidence excerpt (raw statement)	Initial concepts	Subcategory
17	SN271 The Capt. Shorty C had intermittent engine failures (shift from ahead to astern) and a missing 60% of starboard rudder	IC53 Neglecting the management and maintenance of the ship	<i>IS</i> ₁₇ : Ship seaworthiness status not checked
	SN238 The oil tanker FR8 Pride did not test the main engine for stopping/reversing before departure; during navigation, the cooling jacket of cylinder No. 5 cracked, causing the main engine to automatically slow down and lose maneuverability.	IC54 No test performed for parking and reversing the main engine	
	SN258 The NOBRA assigned Manizales to Belmont Anchorage (3 other vessels already anchored) during high water	IC55 Neglecting safety management of ships	
18	SN10 The pilot of the Conti Peridot failed to respond to hydrodynamic forces after meeting the Stolt Span in fog (restricted visibility) and did not inform the approaching Carla Maersk of the handling difficulty.	IC56 In waters with poor visibility	<i>IS</i> ₁₈ : Heavy fog/low visibility
	SN259 The fishing vessel Polaris collided with the tanker Toftevik in the Atlantic Ocean at 1913 (sunset at 1955). It was dusk with fading light, which restricted visibility and affected lookout. The third mate of the Toftevik failed to identify the collision risk in time.	IC57 Early and late shadows affecting visual outlook	
	SN273 In dense sea fog, the master of King Arthur (gas carrier) relied on AIS data and mistakenly judged ANL Wyong (container vessel) as “underway using engine” (actually stopped and drifting due to sea fog). He recklessly altered course to starboard to pass astern of ANL Wyong, leading to a collision between the two vessels.	IC58 Sea fog	
19	SN16 The towing vessel Natures Way Commander was caught in sudden heavy rain (34-knot winds, 40-knot gusts); visibility dropped sharply due to wind and waves, and communication with deckhands was lost, leading to the tow losing control and colliding with moored barges.	IC59 Sudden increase in wind and surge intensity during strong winds and storms	<i>IS</i> ₁₉ : Strong wind
	SN185 Cargo ship Nippon Cargo, when anchored in the Seto Inland Sea affected by Typhoon “Jebi” (Force 15 winds), did not evacuate to a sheltered port in time.	IC60 The impact of a strong typhoon	
20	SN129 The Miraflores Lake area of the Panama Canal had dense vessels (including USCG Cutter Tampa and tugboat Cerro Santiago) and narrow channel (750 feet), with complex navigation environment.	IC61 The navigation environment is more complicated	<i>IS</i> ₂₀ : Complex channel environment
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	SN141 The towing vessel William E Strait encountered a sudden reverse current (speed increased from 1.5 to 2.5 knots) at a Mississippi River bend, with current direction changing from along-channel to transverse, causing the barge flotilla to lose control and deviate to the channel center, colliding with the Margaret Ann.	IC62 Sudden increase in velocity and a change in direction	
21	SN382 Shearwater was not suitable for use as a coastal towing vessel especially through hazardous areas such as the Pentland Firth SN401 Cargo ship German Cargo sailed from Hamburg Port with an unrepaired main engine (the vessel was unfit for navigation).	IC63 Ship is not seaworthy	IS ₂₁ : Equipment not applicable(illegally operated without valid certificates of ship registration and inspection)
	SN247 The insurance of the trawler Jay Dee lapsed in the autumn of 1995, and it did not undergo statutory inspection as required (the last statutory inspection was in August 1993), yet it still illegally engaged in trawling operations near Queensland waters.	IC64 Illegal operation without obtaining the required ship's certificates and inspection	
22	SN155 The Sea Shepherd (fiberglass) had no radar reflector, making it hard to detect.	IC65 No navigational aids are equipped	IS ₂₂ : Inadequate equipment
	SN25 The B.W. Haley was not fitted with AIS;	IC66 No navigational aids	
	SN253 Fog eliminated external visual references (channel banks/aids to navigation). Conti Peridot pilot lost situational awareness; could not judge true heading, leading to disorientation and uncontrolled sheering.	IC67 Cab visual range not satisfactory	
23	SN47 Lifejackets/EPIRB stored in wheelhouse (unreachable).	IC68 Safety management system operation defect	IS ₂₃ : Equipment failure
	SN53 The Kathleen K's starboard engine operated in "astern" but was overpowered by the port engine (ahead); the vessel continued forward and collided with 9 moored recreational vessels.	IC69 Steering engine failure caused the ship to lose control	
	SN26 The Canadian cargo ship Canada Trader encountered steering gear breakdown in the Atlantic Ocean; the hydraulic system leaked, making the rudder completely unresponsive.		

Appendix 2

Collision between container vessel Cosco Hongkong and fish transportation vessel Zhe Ling Yu Yun 135 with loss of 11 lives

At 0218 on 6 March 2011, the UK registered container ship Cosco Hong Kong collided with the China registered fish transportation vessel Zhe Ling Yu Yun 135. The accident occurred in international waters off the coast of Zhejiang Province, China. Zhe Ling Yu Yun 135 sank almost immediately, with the loss of 11 lives. Cosco Hong Kong was not damaged. the performance of Cosco Hong Kong's OOW fell well short of expected standards. He did not correctly apply the collision regulations or follow onboard instructions. Zhe Ling Yu Yun 135 and Cosco Hong Kong both altered their course simultaneously when they were only 1.5 nautical miles apart, with a closing speed exceeding 20 knots. Cosco Hong Kong's OOW was the sole lookout. In darkness, in adverse weather conditions, and among large concentrations of fishing vessels, he did not see that Zhe Ling Yu Yun 135 was approaching. Even though many fishing vessels were concentrated in the area, Cosco Hong Kong's master or OOW did not consider amending the voyage plan, reducing speed, or enhancing the bridge manning.